

Preface to the Updated Edition: What a Book Cannot Learn

Chapter 9 of this book brings a charge against the language model. It calls the model a frozen web of belief: its representations are fixed at the moment training ends, and thereafter no contact with the world propagates back into them. The model can be told it is wrong; it cannot be corrected, because correction, in Quine's sense, is the revision of the web by the world, and the web no longer moves. The model is, in the book's phrase, epistemically static.

I did not, when I wrote that chapter, expect to have to turn it on the book.

But a book is the same kind of object. Its sentences are woven into whatever coherence they have at the moment of writing, and then they stop. The world it set out to describe goes on changing; the description does not. A printed page has no ongoing causal loop to the thing it is about. It cannot register that a claim has gone false, cannot feel the friction when a prediction fails, cannot adjust. A book about the frozenness of machine knowledge is itself a frozen web of belief, and the charge it levels at the model is one it cannot answer on its own behalf. Every book is in this condition. This one merely happens to have described the condition before falling under it.

What you are holding is an updated edition, and the word *updated* deserves to be looked at rather than waved through, because the manner of the updating is exactly the issue. The book did not learn anything. It was corrected from outside. I returned to it; its claims were checked against a world that had moved past the date of composition; the dated ones were patched. In the vocabulary of Chapter 9, this is not the self-correction Quine described but its engineered substitute—the top-down revision by an external hand that the book contrasts, unfavourably, with a web that revises itself. A new edition is to a book what retraining is to a model. The artefact does not update. Someone updates the artefact.

There is a further turn, and it is the one that prompted this preface. A good deal of the checking was done with the help of precisely the kind of system this book is about: a language model, equipped with tools—search, retrieval—that let it hold the book's assertions against the present and flag the ones that no longer held. So a frozen web, the book, was patched against the world by a second frozen web, the model, by means of tools. And the detail worth pausing on is that *neither of them learned*. The book was corrected; the model's parameters did not move either. Two static webs, adjusted to each other and to the present by a process external to both. The tools compensated, after a fashion, for the frozenness of each. They did not thaw either one. The tribunal of experience entered the room, but it entered through neither the book nor the machine.

It entered, as it always does, through the reader. You are the only live web in this arrangement—the only part of the loop that revises in response to the world

as the world actually arrives. The book's frozen sentences become live again only when you hold them against a present they could not see, and notice where they no longer fit. Quine's tribunal does not sit in the text. It sits in the reading.

So, the question that gives this preface its title: how much does the frozenness cost? The honest answer is that it costs unevenly, and that the tools repair the cheaper half. The book's durable claims—the philosophy, the structure of the arguments, the questions held open—were anchored, when written, to things that do not move quickly, and they have not moved. The perishable claims—who holds which post, what a machine costs, what a statute now says—decay on a timescale of months, and those are exactly what this edition went back and patched. That repair is real, and it is worth doing.

But it reaches only what can be checked. A search can confirm a price and catch a dead office-holder. It cannot tell you that an emphasis has gone stale, that a frame has quietly stopped fitting the world, that a judgement which was apt at the time of writing is now the confident description of a country that no longer exists. A book's facts can be kept current while its interpretation rots—and a rotting interpretation reads exactly like a sound one until the reader supplies the friction. That failure, fluency unmoored from a world it no longer tracks, is the very thing Chapter 9 names in the machine. No tool catches it. Only the live web does.

Read this, then, as what it is: a frozen web, patched at its surface against the present, and handed to the one genuinely live web in the transaction—yours. The compensation the tools provide is real but partial and asymmetric. They keep the facts honest. They cannot keep the judgement honest. That work—the only work in any of this that is epistemic in Quine's full sense—has never been the book's to do. It was always yours.

Set Lonnert, updated edition, later that year 2026

Foreword: The Incomprehensible That Arrives

In the summer of 1997 I gave a lecture at a session of the Nordic Summer University, held that year at the University of Reykjavik. The lecture was called *The Technological Condition*. The title echoed Hannah Arendt, but the argument was historical rather than phenomenological. It was an attempt to trace how the technological condition had also become a moral condition: from Francis Bacon's promise that science should be useful "for the benefit and use of life," to Condorcet's stronger claim that the advancement of knowledge would produce moral improvement in man himself, to the twentieth century, where that entailment had been absorbed so thoroughly into society that it had become

difficult to see and harder to evaluate. Georg Henrik von Wright—whose critique of the myth of progress had been central to Scandinavian intellectual life in those years—provided the long perspective: Western technological society is, in historical terms, a very recent and very unusual formation. And one structural observation kept returning: the critique of the technological condition almost always begins from inside the condition and ends there. *Technology is corrected by more technology*. The frame is not questioned; it is adjusted.

That same summer, in New York, Garry Kasparov lost a chess match to IBM's Deep Blue. The loss was disputed, contested, and eventually the subject of a documentary that suggested the machine might have had human help during the decisive game. But the symbolic event had already occurred. The world's greatest chess player, a man who understood the game at a depth no one had matched, had been defeated by a system that did not understand chess at all. It counted. It searched. It found moves Kasparov could not anticipate not because they were deep but because they were inhuman—the product of a kind of deliberation that had no interior, no intuition, no fear of losing.

That summer posed two questions, and this book tries to hold both open. The first is the one New York made concrete: not whether machines can beat us at tasks we value—plainly they can—but what that means for what we call intelligence, for what we call understanding, for the kind of society we are in the process of building around systems whose operations we cannot follow and whose implications we have not begun to settle. The second is the one Reykjavik kept returning to: what kind of condition are we already inside, and from what standpoint can we evaluate it at all?

What makes artificial intelligence unusual as an intellectual subject is that it refuses to stay in any one domain of inquiry. A new pharmaceutical raises biological and ethical questions; a new weapons system raises strategic and moral ones; a new communication technology raises political and cultural ones. AI raises all of these simultaneously, and it adds something the others do not: it directly implicates the nature of mind. To build an artificial intelligence, you have to decide what intelligence is. To align it with human values, you have to decide what values are and whether they can be specified at all. To govern it, you have to decide what governance is for, who it serves, and by what right. These are the oldest questions in philosophy. What is new is that they now have a deployment schedule.

There is a further reason the subject rewards this kind of attention. The attempt to build intelligence turns out to be one of the most revealing ways to understand what intelligence actually is. When engineers specify what a system needs in order to genuinely understand language, they discover requirements—embodiment, ongoing causal engagement with the world, social formation, the capacity for genuine error and correction—that philosophy could only gesture

at from the inside. The system that fails to understand, precisely in its failure, makes visible what understanding requires. This is why the technical chapters of this book are not preliminary to the philosophical ones. They are already doing philosophical work.

The book is organised around three questions, each of which turns out to be harder than it looks.

The first is a technical and scientific question: *What would it take to build genuinely intelligent machines?* The answer, it turns out, depends entirely on what you mean by intelligence—and the debate about that is as live as it has ever been. The first part of the book traces three foundational contributions—von Neumann’s abstract logic of self-replication, Kenneth Stanley’s case against optimisation as a model of discovery, and Yann LeCun’s world-model architecture as a proposed alternative to the dominant paradigm of language models—that together sketch the terrain of the technical debate. (LeCun’s conviction on this point is not merely academic: at the end of 2025 he left the most senior AI post in industry to build the world-model architecture independently, a biographical fact that Chapter 3 takes up.)

The second question is evaluative: *What have we built, and what are its limits?* Part II gathers the critics—the cognitive scientists, AI researchers, and ethicists who have documented what current AI systems cannot do, what the costs of scaling are, and what the most technically serious voices in the field believe we should be worried about. The cautionary voices here—Hinton, Bengio, Russell—are not outsiders sounding alarms. They are among the people most responsible for building what they are warning about. That irony deserves attention.

The third question is philosophical and political: *On whose terms is this being built, and what kind of world does it imply?* Parts III through V take up alignment, practice, consciousness, mythology, non-Western philosophy, embodiment, autonomous systems, and governance. These chapters are the ones that feel most urgent to me, because they are the ones where the questions most directly concern not what machines can do but what we, as societies, have decided to do with them.

A word about what this book is not. It is not a textbook. It is not a complete survey of AI research. It is deliberately selective, following the questions I find most illuminating rather than covering the field. It is also not neutral. It has a point of view: that the discourse around AI has acquired quasi-religious features that warrant scepticism; that the complexity of the alignment problem has been underestimated by optimists and overestimated in certain ways by pessimists; that the non-Western perspectives on intelligence and technology are not merely complementary but genuinely corrective; and that, in Feyerabend’s spirit, epistemic humility is not an obstacle to clarity but its precondition.

The book was written with the assistance of large language models—an irony that is not lost on me, and that Chapter 9 addresses directly. The question of what it means to think alongside a system that processes language without understanding it, that generates arguments it does not hold, that has no stake in being right, is itself one of the most interesting questions the technology raises. I have tried to use it honestly: as a research and editorial tool subordinate to judgment, not as a replacement for it.

Reykjavik is still there. The chess board is still there. The question the summer of 1997 posed is still open. What follows is my attempt to take it seriously.

Set Lonnert, 2026